INCH-POUND

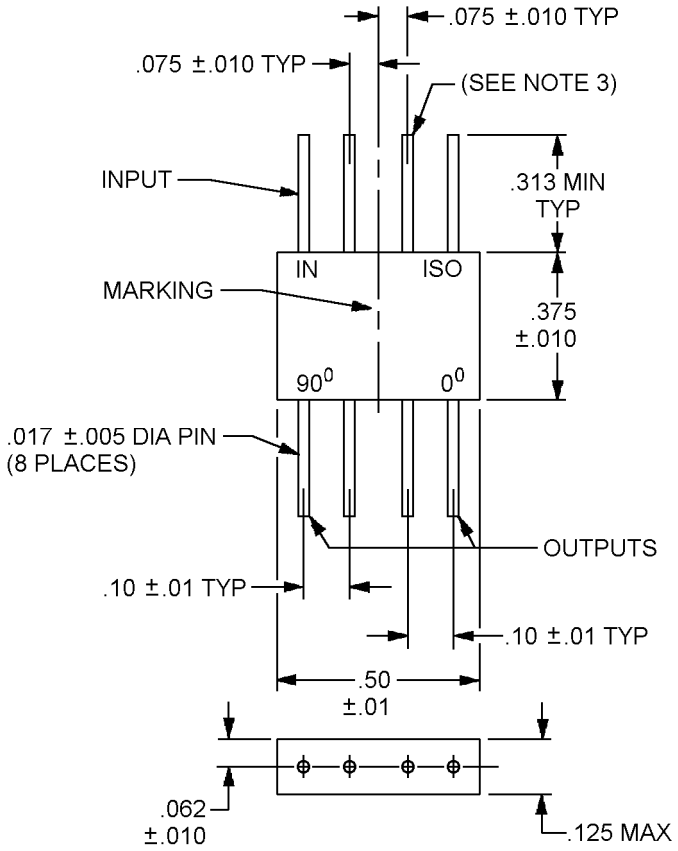
MIL-DTL-23971/8A
w/AMENDMENT 1
19 March 2008
SUPERSEDING
MIL-DTL-23971/8A
21 April 2003

DETAIL SPECIFICATION SHEET
POWER DIVIDERS, QUADRATURE, FLAT PACK

Inactive for new design
after 19 March 2008

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the power divider described herein shall consist of this specification sheet and MIL-DTL-23971.

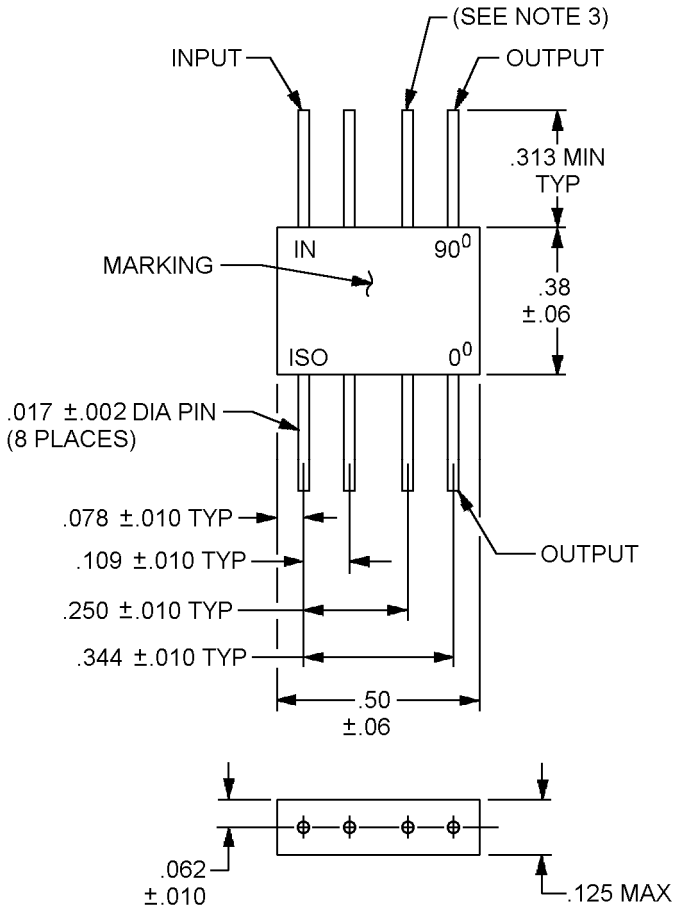


Inches	mm
.005	0.13
.01	0.3
.017	0.43
.062	1.57
.075	1.90
.10	2.5
.125	3.18
.313	7.95
.375	9.52
.50	12.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Pins that are not marked are grounded to the internal circuit of the power divider by the manufacturer.

FIGURE 1. Dimensions and configuration, dash number 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:34Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	.05
.010	.25
.017	.43
.06	1.5
.062	1.57
.078	1.98
.109	2.77
.125	3.18
.250	6.35
.313	7.95
.344	8.74
.38	9.7
.50	12.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Pins not marked are grounded to the internal circuit of the power divider by the manufacturer.

FIGURE 2. Dimensions and configuration, dash number 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:34Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics and physical requirements.

M23971/8-	Impedance (ohms)	Frequency range (MHz)	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance max (degree)	Amplitude balance (dB) max	Power level pk (W)	Weight pounds (grams)	Ambient temperature		Figure
											Operating	Storage	
01	50	7-14	3.0 + 0.2 -0.0	1.2:1	0.5	25	±2.0	±0.75	1/	---	-54°C to +125°C	-54°C to +125°C	1
02	50	40-80	3.0 + 0.2 -0.0	1.2:1	0.5	20	±3.0	±0.75	2/	.062 (28.1)	-55°C to +85°C	-55°C to +85°C	2

1/ Power level: 5 watts to +100°C, derated linearly to 1 watt at +125°C.
2/ Power level: 5 watts at +25°C, derated linearly to 1 watt at +85°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:34Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction: See figures 1 and 2.

Case: Hermetically sealed.

Material: Cold rolled steel 1010.

Finish: Gold plated in accordance with ASTM-B488, type I, code A, class 0.76.

Cover:

Material: Cold rolled steel 1010.

Finish:

Dash number 01: Gold plated in accordance with ASTM-B488, type I, code A, class 0.76.

Dash number 02: Plated electroless nickel in accordance with SAE-AMS-C-26074, class 1, .00012 inch thick.

Terminals:

Dash number 01: Dumet or Kovar, gold plated in accordance with AMS2422.

Dash number 02: Input and output leads: Dumet, type D, in accordance with MIL-STD-1276.

Dash number 02: Ground leads: No. 52 alloy, type F, in accordance with MIL-STD-1276.

Electrical characteristics: See table I.

Weight: See table I.

Ambient temperature: See table I.

Environmental tests: In accordance with MIL-DTL-23971 except:

Dash number 01:

Seal: In accordance with method 112 of MIL-STD-202, test condition C, procedure IIIa, followed by gross leak in accordance with test condition B.

Thermal shock: In accordance with method 107 of MIL-STD-202, test condition B for 10 cycles.

Shock: In accordance with method 213 of MIL-STD-202, test condition I.

Part or Identifying Number (PIN): M23971/8- (dash number from table I).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:34Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-DTL-23971, this specification sheet references the following documents: AMS2422, ASTM-B488, MIL-STD-202, MIL-STD-1276, and SAE-AMS-C-26074.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA CC

(Project 5985-2008-023)

Review activities:
Navy - AS, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:34Z
Check the source to verify that this is the current version before use.